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M3SPCL: Multi-stage multi-grained multi-view supervised prototypical contrastive learning

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ABSTRACT

Most existing multi-view learning methods enhance performance by exploiting view complementarity through multi-stage fusion and enforcing consistency via representation alignment. However, they are largely confined to instance-level modeling and fail to adequately capture category-level semantics, which limits their effectiveness. Recent contrastive learning attempts incorporate category information to improve discriminability, yet they often rely on exhaustive view and sample pairing, leading to prohibitive computational overhead. Consequently, achieving both category-aware effectiveness and computational efficiency remains a fundamental challenge in exploiting dual correlations at the instance and category levels. To this end, we propose M3SPCL, a Multi-stage Multi-grained Multi-view Supervised Prototypical Contrastive Learning framework. In the early stage, M3SPCL adopts a paired view selection strategy to construct contrastive inputs via efficient view concatenation. In the intermediate stage, it integrates supervised and prototypical contrastive learning to enforce fine- and coarse-grained consistency across instance and category levels, where prototype-based matching also reduces dense sample comparisons. In the late stage, an adaptive decision fusion mechanism aggregates view-specific predictions according to their learned contributions. Overall, M3SPCL provides a unified framework that simultaneously enhances effectiveness and efficiency in exploiting dual correlations. Extensive experiments on eight widely used public multi-view benchmark datasets, including image-based datasets (e.g., Handwritten, Scene15, Caltech101) and text-based datasets (e.g., Reuters), demonstrate that M3SPCL consistently outperforms state-of-the-art methods. In particular, M3SPCL achieves improvements of 1.162% in Accuracy, 1.703% in Precision, 1.298% in Recall, and 1.752% in Fscore over competitive baselines, while substantially reducing computational cost and memory usage.

1. Introduction

Multi-view data are ubiquitous in real-world scenarios, where the same object or phenomenon is observed from multiple perspectives or sources. For example, an image can be described by different types of features (Subhashri et al., 2025), and an autonomous driving system perceives its surroundings through multiple sensors, such as cameras and radars. By integrating consistent and complementary information across views, multi-view learning aims to obtain more comprehensive data representations, which are crucial for various downstream tasks such as classification and clustering (Li et al., 2025). Typically, the effective-

ness of the multi-view learning relies on two complementary principles: 1) consistency, which captures shared semantics across views, and 2) complementarity, which emphasizes the view-specific information. In existing multi-view learning literature, these two principles are typically addressed through different modeling mechanisms. Complementarity is often achieved by integrating information at different stages of the learning pipeline, including early, intermediate, and late fusion (Ramachandram & Taylor, 2017). Early fusion methods aim to concatenate raw features from different views into a single joint representation (Chen et al., 2013). Intermediate fusion fuses high-level representations at intermediate network layers to learn a unified joint representation

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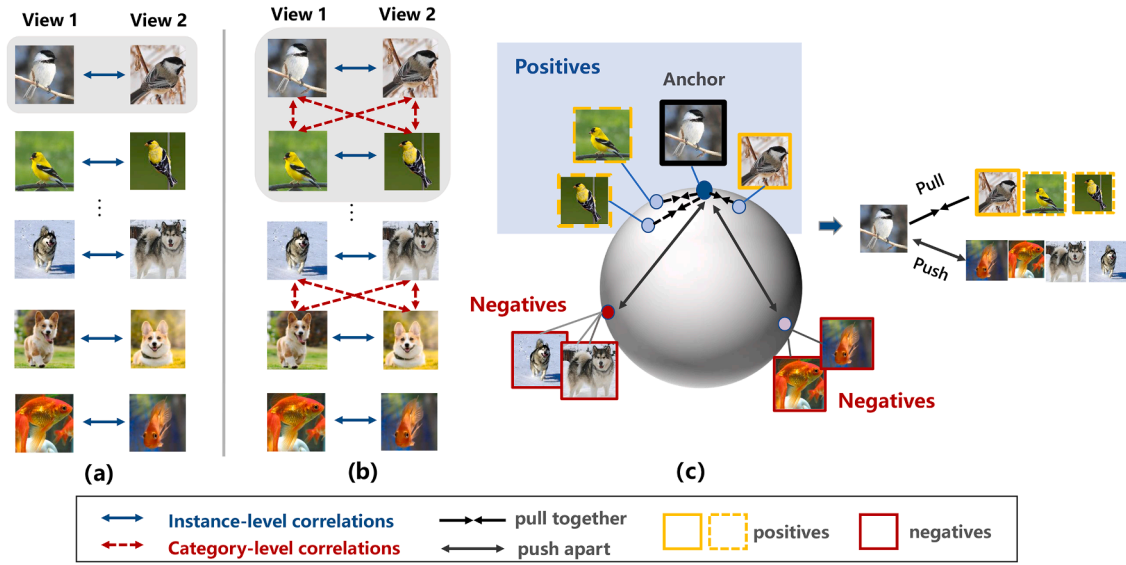


Fig. 1. The dual correlations in multi-view learning. (a) Instance-level correlations; (b) Dual correlations; (c) Category-level consistency in multi-view contrastive learning.

(Li et al., 2018). Late fusion methods operate on independent views and aggregate the outputs of multiple classifiers at the decision level (Zhang et al., 2025). In contrast, consistency is usually enforced by imposing alignment criteria on learned representations, such as maximizing cross-view correlation or similarity (Tang et al., 2023). Representative approaches include correlation-based methods, e.g., canonical correlation analysis (Cai et al., 2026), and contrastive learning (CL) (Tian et al., 2020), which align different views under predefined objectives.

Despite their effectiveness, these multi-stage fusion and criteria-based alignment mechanisms are predominantly designed to model instance-level correlations, i.e., complementarity and consistency across different views of the same sample (as illustrated in Fig. 1 (a)). However, consistency and complementarity are not confined to individual instances. They also naturally exist at the category level, where samples belonging to the same class share common semantic traits while exhibiting diverse characteristics. As depicted in Fig. 1 (b), birds within the same category exhibit common intrinsic traits, such as having two legs, pointed beaks, and the ability to fly, reflecting category-level consistency. Simultaneously, variations in appearance and behavior among different bird breeds highlight their complementarity, enriching the category's diversity. We collectively term the relationships at both the instance and category levels as dual correlations, which provide a unified perspective for exploiting consistency and complementarity in multi-view learning.

Recently, several studies have introduced supervised contrastive learning into multi-view learning to capture category-level semantics (Yang et al., 2021, 2025). As illustrated in Fig. 1(c), these methods not only pull together representations across views of the same instance, but also encourage samples from the same category to be closer in the representation space. Despite these advances, multi-view contrastive learning typically involves exhaustive pairing across views and samples, which becomes even more costly when incorporating category-level supervision. Consequently, existing methods suffer from prohibitive overhead and memory bottlenecks as the number of views or categories increases.

Therefore, existing multi-view contrastive learning methods still struggle to simultaneously achieve strong effectiveness and high efficiency when exploiting dual correlations across instance and category levels. This trade-off naturally leads to two fundamental challenges: (1) **Effectiveness**, how to sufficiently capture both consistency and

complementarity under dual correlations, rather than being restricted to instance-level relationships; and (2) **Efficiency**, how to avoid the prohibitive computational cost caused by exhaustive view and sample pairing when modeling dual correlations with multi-view contrastive learning.

To this end, we propose a novel framework, Multi-stage Multi-grained Multi-view Supervised Prototypical Contrastive Learning (M3SPCL), to address challenges in both effectiveness and efficiency. Fig. 2 demonstrates the overall structure of the model, which jointly exploits consistency and complementarity under dual correlations in a unified manner. Specifically, M3SPCL consists of three tightly coupled modules. Firstly, it constructs contrastive inputs in the paired view selection module by alternately treating each view as the query view, while concatenating the remaining views to form the corresponding key view. This design performs early-stage fusion to expose complementary information, while avoiding exhaustive view pairing. Secondly, based on these constructed view pairs, CL module jointly adopts supervised CL and prototypical CL, encouraging each sample to be close to samples from the same category as well as its corresponding class prototype, while being separated from samples and prototypes of different categories. Importantly, the prototypical formulation not only captures category complementarity in the intermediate stage but also improves efficiency by conducting prototype comparisons. Finally, M3SPCL further evaluates the contribution of each view and aggregates view-specific predictions using adaptive weighting, which realizes late-stage complementarity through a decision ensemble. Through this pipeline, M3SPCL establishes a unified framework that simultaneously mitigates the effectiveness and efficiency challenges. The main contributions are summarized as follows:

- 1) We propose a multi-stage multi-grained multi-view contrastive learning to capitalize on the consistency and complementarity within dual correlations.
- 2) To enhance effectiveness, M3SPCL integrates complementary information through three hierarchical fusion stages while ensuring both fine- and coarse-grained consistency via a dual-level contrastive learning mechanism.
- 3) To improve efficiency, M3SPCL incorporates a paired view selection module to reduce comparison view pairs and employs prototypical contrastive learning to streamline sample comparisons.
- 4) Extensive experiments on benchmark multi-view datasets and

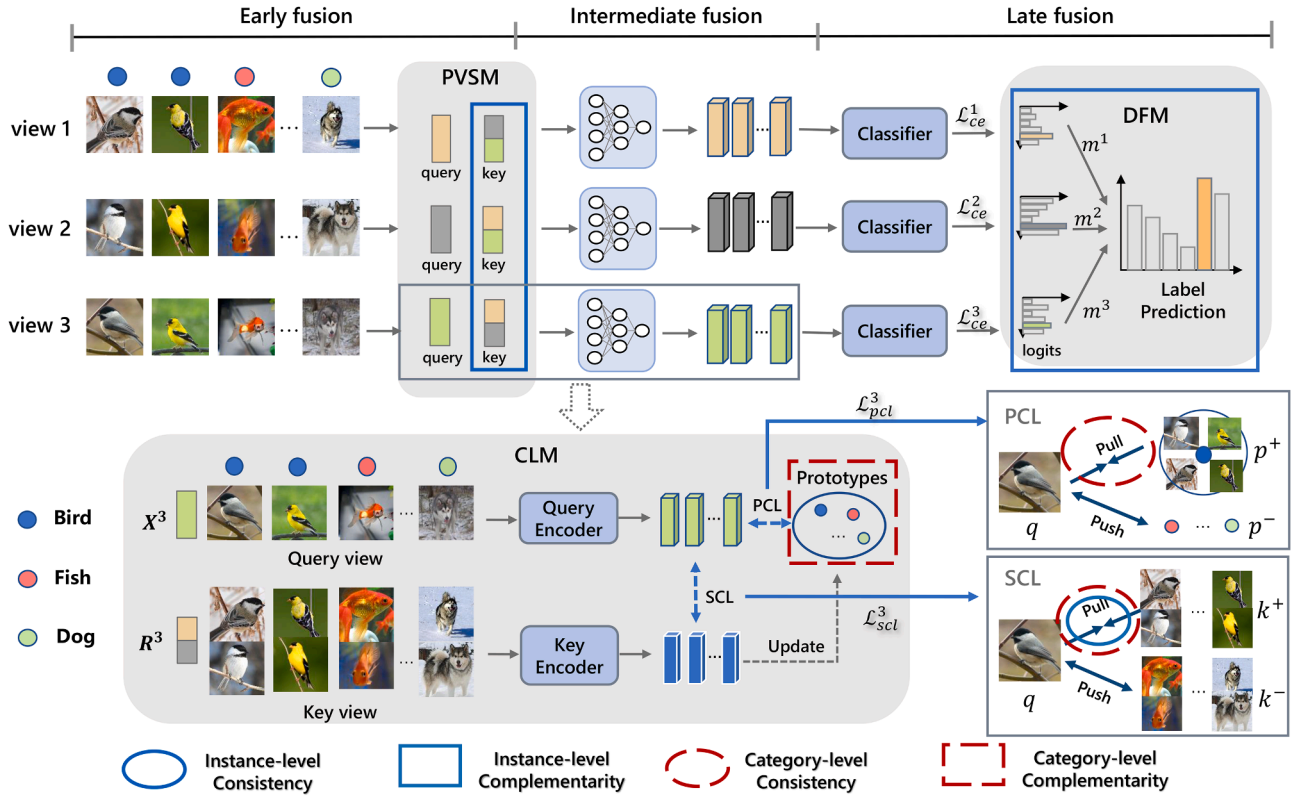


Fig. 2. The pipeline of the M3SPCL (three views as an example). Three modules are seamlessly integrated: 1) Paired View Selection Module (PVSM): Alternates each view as the query view X^v , with the remaining views concatenated to form the key view R^v , $v = 1, 2, 3$. 2) Contrastive Learning Module (CLM): It combines supervised contrastive learning (SCL) and prototypical contrastive learning (PCL). Specifically, the query representation q is attracted to positive matches (keys k^+ and prototype p^+ from the same class) and repelled from negative matches (keys k^- and prototypes p^- from different classes). The prototypes are progressively updated using the key representations. 3) Decision Fusion Module (DFM): The model adaptively adjusts the weights of the logits to generate the final label prediction. By integrating these three modules, M3SPCL not only captures complementary information across early, intermediate, and late stages, but also ensures fine- and coarse-grained consistency through a dual-level contrastive learning mechanism.

detailed ablation studies validate the proposed framework's superior performance and computational efficiency, demonstrating its practicality for multi-view learning tasks.

2. Related work

In this section, we first review related work from the perspective of two fundamental principles in multi-view learning: complementarity and consistency. Subsequently, we categorize existing multi-view contrastive learning into instance-level and category-level paradigms, and discuss their respective strengths and limitations.

2.1. Multi-view learning

Complementarity learning. Most existing multi-view learning methods capture complementarity through fusion at different stages of the learning pipeline. 1) The early (data-level) fusion methods tend to concatenate data from different modalities into a unified single-view representation. Kan et al. (2016) realized that original concatenation can distort the data distribution and exacerbate the curse of dimensionality. To address these limitations, Guo et al. (2014) applied kernel matrices to encode diverse modal information in a multiple kernel learning framework. 2) Intermediate (feature-level) fusion methods first transform each view into a high-level representation via view-specific encoders, and then integrate these at intermediate network layers to learn a joint representation. For example, Tian et al. (2022) incorporated the knowledge distillation into multi-view learning to realize complementarity, which involved treating the view-specific information as the “stu-

dent”. Recently, Li et al. (2024) incorporated instance-based learning to construct an instance attention fusion network for multi-view classification. 3) Late fusion, also known as decision-level fusion, combines decisions or predictions from independently trained classifiers at the final stage of processing (Ramachandram & Taylor, 2017). For instance, Glodek et al. (Xu et al., 2020) designed an adaptive multi-view fusion framework to capture both view-specific information and deep interactive information. A similar adaptive weighted fusion idea was developed in Li et al. (2024), which dug out the complementary information from the enhanced view-specific information. Recently, Han et al. (2022) proposed a trusted multi-view classification model that estimated uncertainty to evaluate instance trustworthiness, integrating multi-view features at the decision level to improve classification reliability.

Consistency learning. Consistency-oriented multi-view learning enforces cross-view agreement by aligning view-specific representations in a shared semantic space. Canonical correlation analysis (CCA) (Kettenring, 1971) represented a foundational approach, identifying linear transformations that maximized view correlations. Kernel-based extensions of CCA (Lian et al., 2020) further captured nonlinear relationships, broadening its applicability to more complex datasets. DeepCCA and its variants (Andrew et al., 2013) extended CCA by using deep neural networks (Cao et al., 2024) to learn complex feature transformations, allowing for the analysis of larger and more intricate datasets. Recent work by Lyu and Fu (2022) strengthened the theoretical foundation of DeepCCA, demonstrating its ability to eliminate unknown distortions and identify shared information across modalities, even under finite sample conditions. Multi-view Discriminant Analysis (MvDA) (Kan et al., 2016) leveraged label information to learn a shared discriminant space

by maximizing inter-class variance while minimizing intra-class variance. Nonetheless, these methods are often limited by linear assumptions and scalability. Recently, contrastive learning has emerged as a more flexible and effective paradigm for modeling multi-view consistency in deep representation learning (Chen et al., 2023; Tian et al., 2020).

Typically, existing multi-view learning methods implement these principles through multi-stage fusion to exploit complementarity and view alignment to enforce consistency, both predominantly at the sample level. In other words, they mainly focus on correlations among different views of the same instance (i.e., instance-level correlations), while neglecting correlations that may exist at the category level. This gap highlights the need for new multi-view frameworks that jointly model dual correlations at both the instance and category levels.

2.2. Multi-view contrastive learning

Instance-level CL. Multi-view contrastive learning enforces alignment by contrasting positive and negative sample pairs across views, which serves as a prevalent consistency learning method. Early multi-view contrastive methods focus on instance-level consistency, where only different views of the same instance are pulled together in the representation space. Representative examples include contrastive multiview coding (Tian et al., 2020) and its extensions in multi-view clustering. For example, Xu et al. (2022) proposed a multi-level, fusion-free multi-view clustering method to balance semantic consistency and view-private reconstruction. Ke et al. (2021) introduced a contrastive fusion network based on the information bottleneck theory, enhancing multi-view alignment while preserving view-specific information. Furthermore, Sun et al. (2024) combined CL with a discriminator and imposed orthogonal constraints on specific features to satisfy the consistent and complementary principles. In this way, only different views of the same instance are treated as positive pairs, whereas all other samples, including those from the same category, are regarded as negatives. Such instance-level contrastive learning inevitably introduces false negatives, which limits its ability to capture category-level semantics.

Category-level CL. To alleviate the false negative issue, several multi-view CL approaches introduce category-level supervision, either through supervised CL or pseudo-label-driven mechanisms. Supervised CL mitigates false negatives by defining positive and negative pairs with labels. For instance, Yang et al. (2021) proposed a noise-robust contrastive loss that adaptively down-weights the influence of unreliable negative pairs during optimization, preventing potential false negatives from dominating representation learning. Lin et al. (2022) unified cross-view consistency learning and missing-view recovery by combining instance-level and category-level contrastive learning with dual prediction. Under a supervised setting, the category-level contrastive objective implicitly alleviates false negatives while enforcing semantic consistency across views. In unsupervised learning, pseudo-label relationships are often constructed based on representation similarity or structural cues, thereby implicitly mitigating false negatives. Yan et al. (2023) introduced structure-guided contrastive learning to preserve sample-wise structural relationships when aligning consensus and view-specific representations. Structural relationships inferred from the consensus representation serve as implicit pseudo-labels, thereby alleviating false negatives in unsupervised multi-view clustering. Similarly, a dual contrastive calibration network was introduced by Yang et al. (2023), which aligns view similarity graphs with high-confidence pseudo-label graphs to enforce cross-view semantic consistency among similar samples. Furthermore, Yan et al. (2024) introduced an anchor-sharing and clusterwise contrastive learning, where cluster-level transition probabilities serve as soft pseudo-labels. By down-weighting negative pairs with high cluster affinity, the method effectively alleviates false negatives in multi-view contrastive learning.

Despite their effectiveness in alleviating false negatives by incorporating the category semantics, these methods still exhibit notable lim-

itations. First, although category-level supervision is introduced, many existing methods still primarily focus on enforcing cross-view consistency, while complementary information across different views is insufficiently exploited. Second, incorporating category-level supervision through supervised labels or pseudo-labels often involves pairing multiple views across many samples from the same category, which introduces significant computational and memory costs. Consequently, how to jointly leverage consistency and complementarity under dual correlations, while maintaining computational and memory efficiency, remains an open challenge.

3. Methodology

This section presents the overall framework of M3SPCL, which consists of three tightly coupled modules, as illustrated in Fig. 2. In the early stage, a paired view selection module is designed to construct contrastive inputs by alternately treating each view as the query view and concatenating the remaining views as the key view, enabling efficient early-stage fusion. In the intermediate stage, a contrastive learning module integrates supervised contrastive learning and prototypical contrastive learning to jointly enforce fine-grained and coarse-grained consistency while capturing category-level complementary information. In the late stage, a decision fusion module aggregates view-specific predictions to further exploit complementary information at the decision level. Together, these modules enable M3SPCL to effectively exploit dual correlations while maintaining computational efficiency.

Problem Statement: Given a labeled multi-view dataset $(\mathcal{X}, Y)$, we denote the multi-view observations as $\mathcal{X} = \{X^1, X^2, \dots, X^V\}$, where V is the number of views. For the v -th view, $X^v \in \mathbb{R}^{N \times D^v}$ represents the corresponding data matrix, with N samples and feature dimension D^v . Specifically, the i -th sample in view v is denoted by x_i^v , which corresponds to the i -th row of X^v . The associated label set is $Y = \{y_i\}_{i=1}^N$, where $y_i \in \{1, \dots, C\}$ denotes the class label of the i -th sample and C is the number of classes. Other notations used throughout this paper are summarized in Table 1.

3.1. Paired view selection module

In the early stage, we design a paired view selection module to generate two inputs (query and key view) for the subsequent siamese network to perform contrastive learning. Specifically, every view takes turns to be the query view while the others are concatenated as the key view.

In each iteration v , the raw data X^v is designated as the query view, while the remaining views are concatenated to form a joint representation R^v , referred to as the key view:

$$R^v = f_{\text{concate}}^{v, u \neq v}(X^v), \quad (1)$$

where $f_{\text{concate}}(\cdot)$ denotes the concatenation operation.

The query-key pairs $\{X^v, R^v\}_{v=1}^V$ are then passed into contrastive learning modules for training. Subsequently, the query encoders $f_Q^v(\cdot)$ and key encoders $f_K^v(\cdot)$ are used to extract the query representations $Q^v = \{q_1^v, q_2^v, \dots, q_N^v\}$ and key representations $K^v = \{k_1^v, k_2^v, \dots, k_N^v\}$, respectively. Let x_i^v and r_i^v denote the i -th sample of the input X^v and R^v , we have

$$q_i^v = f_Q(x_i^v), \quad k_i^v = f_K(r_i^v). \quad (2)$$

For simplicity, both q_i^v and k_i^v are normalized using L_2 normalization.

Innovation for efficiency. Existing multi-view CL methods handle more than two views through two paradigms: core view and full graph (Lin et al., 2022; Tian et al., 2020). In Fig. 3, we use $V = 4$ as an illustrative example. The core view paradigm selects one central view (e.g., v_1) and models its interactions with the remaining views, i.e., $(v_1, v_2), (v_1, v_3), (v_1, v_4)$. While this strategy reduces computational cost, it inevitably ignores correlations among the non-central views, such as

Table 1
Basic notations used in the method.

Notation	Meaning
X^v	Data matrix of the v -th view, where x_i^v denotes the i -th sample
R^v	Key view corresponding to the query view X^v , with r_i^v denoting the i -th sample
Y	Ground-truth labels, where y_i is the label of the i -th sample
N, V, C	Number of samples, views and classes
D^v	Feature dimension of the v -th view
d	Query, key, prototypes representation dimensions.
Q^v	Query representations encoded from the v -th view, with q_i^v representing the i -th sample
K^v	Key representations corresponding to the v -th view, with k_i^v representing the i -th sample
Pos_i^v	Set of positive keys associated with anchor q_i^v
All_i^v	Set of all keys (including both positive and negative) associated with anchor q_i^v
P^v	Class prototype sets of the v -th view, where p_c^v represents prototype of the c -th class in the v -th view
S^v	Logits for the v -th view, with s_i^v representing the i -th sample
m^v	Weight assigned to the v -th view in decision fusion
τ	Temperature parameter of the contrastive learning module
μ	Momentum coefficient for prototype updating
α	Balance parameter in total loss
$f_Q^v(\cdot), f_K^v(\cdot), g^v(\cdot)$	Query encoder, key encoder and classifier of the v -th view

Paradigm ($V=4$)	(a) Core View	(b) Full Graph	(c) Our
Connection way			
View pairs			
Number of CL operations	$2 \times (V - 1) = 6$	$2 \times C_V^2 = 12$	$V = 4$

Fig. 3. Graphical diagrams of view pair selection in multi-view CL (Suppose $V = 4$). The edges denote correlations considered in the corresponding paradigm. In view pairs, the first row represents the query view, and the second row denotes the key view. (a) Core view: it focuses on partial interactions, reducing complexity but missing some correlations. (b) Full graph: it considers all correlations but with increasing computational costs. (c) Our paradigm balances completeness and efficiency.

$(v_2, v_3), (v_2, v_4), (v_3, v_4)$, leading to incomplete modeling of cross-view relationships. As a more general formulation, the full graph paradigm considers all possible view pairs but is computationally expensive shown in Fig. 3 (b). Furthermore, both paradigms change the query and key positions for each paired view. As a result, the core view and full graph paradigms need to conduct the CL between views for 6 and 12 times, respectively.

Our proposed paired view selection module, illustrated in Fig. 3 (c), preserves essential correlation information while reducing computational overhead. By rotating the query role across views and aggregating all remaining views into a unified key, our strategy implicitly models all pairwise cross-view correlations while eliminating the need for explicit pair enumeration and query-key swapping. Notably, our method performs CL only V times, whereas the core view requires $2(V - 1)$ times and the full graph needs $2C_V^2$ times. When $V = 2$, all paradigms have the same number of CL operations, but when $V > 2$, our design is more efficient.

Generally, this design not only captures the complementarity via early original concatenation but also reduces computational complexity by avoiding the need for exhaustive pairwise combinations across all views.

3.2. Contrastive learning module

In the intermediate stage, supervised and prototypical CL are integrated within a unified module to achieve both fine- and coarse-

grained consistency. Complementary information is effectively captured through prototype integration. By leveraging prototypical CL, the framework eliminates the necessity for exhaustive pairwise comparisons, thereby significantly reducing computational overhead.

Supervised CL. Supervised CL involves maximizing the similarity between positive pairs (samples from the same class, i.e., $\{q_i^v, k_p^v\}$) and minimizing the similarity between negative pairs (samples from different classes, i.e., $\{q_i^v, k_a^v\}, a \neq p$). By doing this, the model ensures category-level consistency in a fine way. The supervised contrastive loss is formulated as follows:

$$\mathcal{L}_{scv}^v = -\frac{1}{N} \sum_{i=1}^N \frac{1}{|Pos_i^v|} \sum_{p \in Pos_i^v} \log \frac{\exp(q_i^v \cdot k_p^v / \tau)}{\sum_{a \in All_i^v} \exp(q_i^v \cdot k_a^v / \tau)}, \quad (3)$$

where query feature q_i^v serves as the anchor, and k_p^v denotes the positive matches corresponding to the anchor q_i^v . Pos_i^v and All_i^v are the sets of positive keys and the entire set of keys (including both positive and negative samples) associated with the anchor, respectively. $|Pos_i^v|$ denotes the number of positive keys. $\tau > 0$ is the temperature parameter that controls the scaling of similarity in contrastive learning.

Supervised CL not only pulls together representations of multiple views from the same sample, but also aggregates samples belonging to the same category. This enables M3SPCL to explicitly model category-level consistency, which cannot be adequately captured by instance-level alignment alone.

Prototypical CL. During training, prototypical CL encourages the anchor q_i^v to be closer to its corresponding class prototype p_c^v while distancing it from prototypes of other classes. Each anchor has a single positive prototype and $C - 1$ negative prototypes. This mechanism enhances category-level consistency in a coarse-grained manner. The prototypical contrastive loss is defined as:

$$\mathcal{L}_{pcl}^v = -\frac{1}{N} \sum_{i=1}^N \log \left\{ \frac{\exp(q_i^v \cdot p_c^v / \tau)}{\sum_{j=1}^C \exp(q_i^v \cdot p_j^v / \tau)} \right\}, \quad (4)$$

where p_c^v denotes the positive prototype of the anchor q_i^v , which shares the same label c as the anchor (i.e., $y_i = c$). The temperature parameter τ matches that in $\mathcal{L}_{scv}$.

Prototypes updating. Instead of using a momentum encoder (He et al., 2020a), we implement cross-batch dynamic prototypes to store and retrieve information efficiently. Specifically, the prototypes are initialized as zero vectors. In each batch, prototypes are updated from current-batch keys using a momentum-based moving average technique, progressively incorporating information from preceding batches. Let $\hat{k}_c^v$ denotes the average value of the key representations for the c -th class in the current batch of the v -th view, and $p_c^{v'}$ represents the prototype for

the c -th class in the v -th view from the preceding batch. Then the class prototype is updated according to

$$p_c^v = \text{Normalize} \left(\mu p_c^{v'} + (1 - \mu) \hat{k}_c^v \right), c = 1, 2, \dots, C, \quad (5)$$

where the momentum coefficient $\mu \in (0, 1)$ controls the balance between the historical prototype and the current batch statistics. Through the updating of the prototypes, the model guarantees the category-level complementarity via feature integration.

Prototypes for efficiency. Multi-view and label guidance provide a large number of view pairs and available positive-negative comparisons (Tian et al., 2020), which increases computational overhead (He et al., 2020b; Khosla et al., 2020). To address the above issues, we design a CL module that combines supervised CL with prototypical CL in an end-to-end framework. To illustrate how prototypes alleviate the efficiency bottleneck, we compare three scenarios: 1) Scenario 1 represents multi-view supervised CL operating with a fixed batch size. 2) Scenario 2 demonstrates the batch-expanded supervised CL for enhanced performance. 3) Scenario 3 presents our proposed approach, which incorporates prototypical CL with supervised CL without batch expansion.

Example 1. Suppose the data size $n \gg N$, batch size $N > C$, and the number of class C is 10.

- **Scenario 1 (supervised CL):** If the batch size $N = 128$, and the query q_i^v has 5 positives ($|P(x_i)| = 5$). During the training process, the query q_i^v can only contrast the samples within the current batch. In this case, the number of positives and negatives of q_i^v are 5 and 123, respectively. This results in a memory consumption linear to the product of the positive and negative sizes. Hence, the number of comparisons performed is $5 \times 123 = 615$.
- **Scenario 2 (supervised CL with expanded batch):** Good representations highly depend on a large number of comparisons. Enlarging the batch size is an intuitive solution but introduces more computational burden. Assuming we enlarge the batch size to $N = 256$, the query q_i^v might have more positive samples, i.e., $|P(x_i)| = 6$. In this case, the number of positives and negatives of q_i^v are 6 and 250, respectively, resulting in $6 \times 250 = 1500$ comparisons in Formula (3).
- **Scenario 3 (Our supervised + prototypical CL):** To incorporate more comparative information without significantly increasing the computational burden, we introduce the prototypes to preserve the information from all previous batches. Each query has only 1 positive prototype and 9 negative prototypes. Suppose the batch size remains at 128, the query q_i^v now has 6 positive matches (5 individual positives and 1 positive prototype) and 132 negatives (123 individual negatives and 9 negative prototypes) in Formula (6). The number of comparisons performed is $6 \times 132 = 792$.

It can be observed that $615 < 792 < 1500$. This innovation ensures that we maintain a rich representation of the data while reducing the number of comparisons needed, thereby alleviating computational pressure without loss of information quality.

Combination. The combination of supervised CL and prototypical CL forms a dual-level contrastive learning mechanism, harnessing both consistency and complementary information effectively, while alleviating the efficiency limitations of earlier contrastive learning approaches. Consequently, CL is performed on a query and N individual keys as well as C prototypes. The individual keys retain current batch information in a fine-grain way, while the prototypes preserve coarse information from previous batches. The dependency on a large number of individual keys is relieved, as prototypes provide substantial comparative information.

By balancing the strengths of supervised CL and prototypical CL, as expressed in Formula (6), the proposed framework not only captures data commonality and diversity but also effectively mitigates computational overhead. Formally,

$$\mathcal{L}_{cl}^v = \mathcal{L}_{sc}^v + \mathcal{L}_{pcl}^v. \quad (6)$$

3.3. Decision fusion module

In the late stage, the decision fusion module aggregates predictions from multiple classifiers, enabling the integration of complementary information at the decision level.

Classifier learning. For the v -th view, a view-specific classifier $g^v(\cdot)$ is trained using the high-quality representation q_i^v . Let $s_i^v = g^v(q_i^v)$ represents the logits for the i -th sample, the logits $S^v = \{s_1^v, s_2^v, \dots, s_N^v\}$ can be obtained from the classifiers. The cross-entropy loss for the view-specific classifier is formulated as:

$$\mathcal{L}_{ce}^v = - \sum_{i=1}^N y_i \log s_i^v, \quad (7)$$

To ensure compatibility between representation learning and classifier learning, these components are unified within a single framework. The total loss for the v -th view is defined as:

$$\mathcal{L}^v = \mathcal{L}_{ce}^v + \alpha \mathcal{L}_{cl}^v, \quad (8)$$

where $\mathcal{L}_{cl}^v$ is the contrastive loss and α is a balance parameter.

Adaptive decision fusion. Each view captures unique aspects of the data, and the decision fusion module leverages this diversity to enhance classification performance. Inspired by Xu et al. (2020), an adaptive weighting strategy is employed to dynamically adjust the contribution of each view during training based on its relevance to the classification task. The total loss is formulated as:

$$\mathcal{L}_{total} = \sum_{v=1}^V (m^v)^\gamma \mathcal{L}^v(Q, K, P, Y, S), \quad (9)$$

where m^v represents the weight of the v -th view, and $\gamma > 1$ is a power exponent parameter that regulates the distribution of weights. For simplicity, $\mathcal{L}^v(Q, K, P, Y, S)$ means $\mathcal{L}(Q^v, K^v, P^v, Y, S^v)$, where Q^v, K^v, P^v, Y , and S^v denote the query, key, prototypes, labels, and logits matrices of the v -th view, respectively.

To prevent trivial solutions (e.g., assigning all weights to a single view), the power operation is introduced to refine the distribution of weights across views. The optimization problem can thus be expressed as:

$$\min_{\{m^v\}_{v=1}^V} \left(\sum_{v=1}^V (m^v)^\gamma \mathcal{L}^v(Q, K, P, Y, S) \right) \text{ s.t. } \sum_{v=1}^V m^v = 1. \quad (10)$$

This formulation ensures that the weights are dynamically adjusted to balance the contributions of all views, avoiding over-reliance on a single view and optimizing the joint classification performance.

Subsequently, we employ an alternating optimization strategy to update the network parameters and view weights iteratively. In each iteration, the encoder parameters $\{f_Q^v(\cdot), f_K^v(\cdot), g^v(\cdot)\}_{v=1}^V$ are updated while keeping the weights $\{m^v\}_{v=1}^V$ fixed. Subsequently, with the network parameters frozen, the weights $\{m^v\}_{v=1}^V$ are updated as:

$$m^v = \frac{(\mathcal{L}^v)^{\frac{1}{1-\gamma}}}{\sum_{l=1}^V (\mathcal{L}^l)^{\frac{1}{1-\gamma}}}. \quad (11)$$

By iteratively refining the parameters and weights, this module effectively integrates multi-view diversity and ensures robust decision fusion. Further implementation details are provided in Appendix A.

In general, our model M3SPCL can offer an effective and efficient way to multi-view learning. For effectiveness, the proposed method integrates complementary information in three stages: original view concatenation for early fusion, prototype incorporation for intermediate fusion, and multi-view decision ensemble for late fusion. Furthermore, the CL module ensures category consistency through individual and prototype comparisons at both fine and coarse levels. For efficiency, the design of paired view selection and the prototypes reduces the computational consumption.

3.4. Complexity analysis

We analyze the per-mini-batch complexity of M3SPCL with batch size n , number of views V , number of classes C , and embedding dimension d . In the paired view selection module, constructing R^v by concatenating the remaining $V-1$ views incurs $\mathcal{O}(n \sum_{u \neq v} D^u)$ for each view, resulting in $\mathcal{O}(n(V-1) \sum_{u=1}^V D^u)$ overall. In the contrastive learning module, for each view, supervised CL requires computing an $n \times n$ similarity matrix with complexity $\mathcal{O}(n^2 d)$, while prototypical CL compares n queries with C prototypes, yielding $\mathcal{O}(nC d)$. Prototype updating is linear, costing $\mathcal{O}(nd + C d)$ per view by accumulating class-wise means and applying momentum updates. Thus, the total complexity of this module is $\mathcal{O}(V n^2 d + V n C d + V n d + V C d)$. The decision fusion module is lightweight, with view-wise cross-entropy loss computation costing $\mathcal{O}(V n C)$ and fusion weight updating costing $\mathcal{O}(V)$. Combining all components, the overall per-mini-batch complexity of M3SPCL is $\mathcal{O}(n(V-1) \sum_{u=1}^V D^u + V n^2 d + V n C d + V n d + V C d + V n C + V)$. In practice, since $V, C \ll n$, the complexity is dominated by the supervised contrastive term $\mathcal{O}(V n^2 d)$.

4. Why does the CL module contribute to multi-view learning?

Intrinsic properties of the CL. In multi-view learning, the consistency and complementarity principles play a central role in effectively leveraging multi-view information. These principles align closely with two key properties of CL: alignment and uniformity, which are known to enhance representation learning (Wang & Isola, 2020). Specifically, alignment ensures that representations of positives share common information and preserve invariant characteristics, while uniformity helps distribute representations across the feature space to capture diverse information. To further elucidate this connection, we conduct a theoretical analysis of the correlations between the alignment and uniformity properties in CL and the consistency and complementarity principles in multi-view learning. To begin with, we provide the definition of alignment and uniformity.

Definition 1. (Alignment). The representations of positive pairs are encouraged to be similar. We say q and k is perfectly aligned if $q \cdot k = 1$, a.s., over $(q, k) \sim p_{\text{pos}}$, p_{pos} is the distribution.

Definition 2. (Uniformity). The distribution of representations over the unit hypersphere should be uniform, which can preserve the specific information as much as possible.

Connecting to the CL module in M3SPCL. As mentioned above, the CL module in M3SPCL excavates consistent and complementary information among the views. For each paired view, we can derive

$$\begin{aligned}
 \mathcal{L}_{cl}^v &= \mathcal{L}_{sc}^v + \mathcal{L}_{pc}^v \\
 &= -\frac{1}{N} \sum_{i=1}^N \frac{1}{|P(x_i)|} \sum_{p \in P(x_i)} \log \frac{\exp(q_i^v \cdot k_p^v / \tau)}{\sum_{a \in A(x_i)} \exp(q_i^v \cdot k_a^v / \tau)} \\
 &\quad - \frac{1}{N} \sum_{i=1}^N \log \frac{\exp(q_i^v \cdot p_c^v / \tau)}{\sum_{j=1}^C \exp(q_i^v \cdot p_j^v / \tau)} \\
 &= -\frac{1}{N} \sum_{i=1}^N \frac{1}{|P(x_i)|} \sum_{p \in P(x_i)} \left(\frac{q_i^v \cdot k_p^v}{\tau} \right) + \frac{1}{N} \sum_{i=1}^N \log \sum_{a \in A(x_i)} \exp \left(\frac{q_i^v \cdot k_a^v}{\tau} \right) \\
 &\quad - \frac{1}{N} \sum_{i=1}^N \left(\frac{q_i^v \cdot p_c^v}{\tau} - \log \sum_{j=1}^C \exp \left(\frac{q_i^v \cdot p_j^v}{\tau} \right) \right) \\
 &= -\frac{1}{N \tau} \sum_{i=1}^N \left(\frac{\sum_{p \in P(x_i)} q_i^v \cdot k_p^v}{|P(x_i)|} + q_i^v \cdot p_c^v \right) + \frac{1}{N} \sum_{i=1}^N (\log A_i^v + \log B_i^v) \\
 &= \underbrace{-\frac{1}{N \tau} \sum_{i=1}^N q_i^v \cdot \left(\frac{\sum_{p \in P(x_i)} k_p^v}{|P(x_i)|} + p_c^v \right)}_{(a)} + \underbrace{\frac{1}{N} \log \prod_{i=1}^N A_i^v B_i^v}_{(b)},
 \end{aligned}$$

where the A_i^v and B_i^v are defined as:

$$A_i^v = \sum_{a \in A(x_i)} \exp(q_i^v \cdot k_a^v / \tau), \quad B_i^v = \sum_{j=1}^C \exp(q_i^v \cdot p_j^v / \tau). \quad (12)$$

For the fixed $\tau > 0$, there are some findings:

1) The proposed loss function comprises two components: term (a) and term (b). As proven in Appendix B, term (b) has a lower bound, which simplifies the optimization objective to minimize term (a). Removing the constants N and τ from term (a), the objective can be reformulated as maximizing $\sum_{i=1}^N q_i^v \cdot \left(\frac{\sum_{p \in P(x_i)} k_p^v}{|P(x_i)|} + p_c^v \right)$. This formulation ensures consistent alignment of positive pairs, reinforcing the alignment property in CL. In the context of multi-view learning, this alignment directly supports the consistency principle by producing similar representations for positive pairs across different views. Conversely, the consistency achieved in multi-view learning further enhances the alignment property in CL.

2) Extremely, suppose the positive pairs in both keys and prototypes are perfectly aligned (i.e., their inner product equals 1, as defined in Definition 1). Under this condition, term (a) simplifies to a constant value of $-\frac{2}{\tau}$. Minimizing the loss then becomes equivalent to minimizing the term $\sum_{i=1}^N \left(\sum_{a \in A(x_i)} q_i^v \cdot k_a^v + \sum_{j=1}^C q_i^v \cdot p_j^v \right)$. In this term, $A(x_i)$ includes both positive and negative pairs. For positive pairs, the inner product remains fixed at 1 due to perfect alignment. Therefore, minimizing this term primarily emphasizes reducing the similarity between queries and negative pairs. This process ensures that negative samples are pushed further apart in the embedding space. By doing so, the term promotes uniformity in the learned representations, spreading them across the embedding space and reducing redundancy. Consequently, this enhances the ability of multi-view learning to preserve view-specific information.

In summary, the first term (a) is the alignment term, promoting consistent representations across views, while the second term (b) is the uniformity term, preserving independent information from each view to avoid redundancy. Notably, while alignment and uniformity are not synonymous with the two principles, both properties contribute to capturing consistent and complementary information in multi-view learning.

5. Experiments

In this section, extensive experiments are employed to validate the effectiveness and efficiency of the proposed method. All the experiments are conducted on Pytorch 1.10.2 and the Ubuntu 18.04 OS with NVIDIA 2080Ti GPU. The source code is available at <https://github.com/Mironly33/M3SPCL>.

5.1. Datasets description

We compare M3SPCL and benchmark methods on eight public multi-view datasets. 1) *Handwritten*: It consists of six-view features for handwritten digits from 0 to 9., each with 200 patterns, totaling 2000 patterns. 2) *LandUse21* (Yang & Newsam, 2010): This dataset comprises 2100 satellite images, each represented by a three-view feature set, across 21 land categories. 3) *BDGP* (Cai et al., 2012): It is derived from Drosophila embryo images, comprising 2500 samples across five object classes, where each sample is described by features from two distinct views. 4) *Scene15* (Fei-Fei & Perona, 2005): It comprises 4485 images, each represented by three distinct views, covering 15 categories of indoor and outdoor scenes. 5) *Caltech101* (Li et al., 2015): This dataset includes 9144 images captured from six different views, covering 101 object categories, along with an additional background category. 6) *Fashion* (Xiao et al., 2017): This dataset comprises 10,000 images across 10 product categories. Each image is represented in three distinct views. 7) *Reuters* (Amini et al., 2009): It consists of 18,758 samples across six

Table 2
Details of the experimental datasets.

Datasets	# Samples (N)	# Views (V)	# Classes (C)	# Feature Dimensions
Handwritten	2000	6	10	240 + 76 + 216 + 47 + 64 + 6
LandUse21	2100	3	21	20 + 59 + 40
BDGP	2500	2	5	1,750 + 79
Scene15	4485	3	15	20 + 59 + 40
Caltech101	9144	6	102	48 + 40 + 254 + 1,984 + 512 + 928
Fashion	10,000	3	10	784 + 784 + 784
Reuters	18,758	5	6	21,531 + 24,892 + 34,251 + 15,506 + 11,547
NoisyMNIST	30,000	2	10	784 + 784

Table 3
Description of the compared methods.

Methods	Fusion strategies	Principles	Correlations	CL paradigm
DeepCon	Early fusion	Complementarity-only	Instance-level	–
MvCLN	Intermediate fusion	consistency-only	Category-level	Core view
DCP	Intermediate fusion	consistency-only	Dual	Full graph
TMC	Late fusion	Complementarity-only	Instance-level	–
ECML	Late fusion	Complementarity-only	Instance-level	–
MTS-NET	Intermediate fusion	Both principles	Instance-level	–
MvNNcor	Late fusion	Both principles	Instance-level	–
MvESR	Intermediate and late fusion	Both principles	Instance-level	–
M3SPCL	All applied	Both principles	Dual	Paired view selection

classes, with each sample represented by five languages. 8) *NoisyMNIST* (Wang et al., 2015): This dataset is a two-view version of MNIST with 70,000 samples across 10 classes. We randomly selected 30,000 instances for our experiments. Details of these datasets are described in Table 2.

5.2. Compared methods

In this section, we compare M3SPCL with seven state-of-the-art methods. Table 3 provides a clear comparison between these methods and our model.

DeepCon: It is an early fusion method that concatenates all views as input to a DNN for feature learning.

MvCLN (Yang et al., 2021): It formulates the view alignment problem within a category identification framework. By leveraging a noise-robust contrastive loss, it effectively mitigates the impact of noisy labels and achieves category alignment. Since MvCLN is primarily designed for two views, we evaluated all pairwise combinations of views and used the best result for comparison.

DCP (Lin et al., 2022): It combines the multi-view consistency learning and missing view recovery into a unified framework. Although it is designed for data-missing scenarios, it can adapt to complete-view learning with an appropriate selection of the missing ratio.

TMC (Han et al., 2022): TMC is a robust evidence-level multi-view late fusion integration that considers the trustworthiness of different views.

ECML (Xu et al., 2024): ECML is a method that handles conflicting information in multi-view learning by combining decision results and reliability from different views to make reliable predictions.

MTS-NET (Tian et al., 2022): It incorporates knowledge distillation into multi-view classification to thoroughly leverage the view consistency and discrepancy.

MvNNcor (Xu et al., 2020): MvNNcor integrates view-specific and deep interactive information, and employs a novel multi-view fusion strategy for joint decision-making.

MvESR (Hao et al., 2023): MvESR leverages cross-view interactions to enhance view-specific information, and blends cross-view consistency with adaptive-weighting decision fusion for multi-view classification.

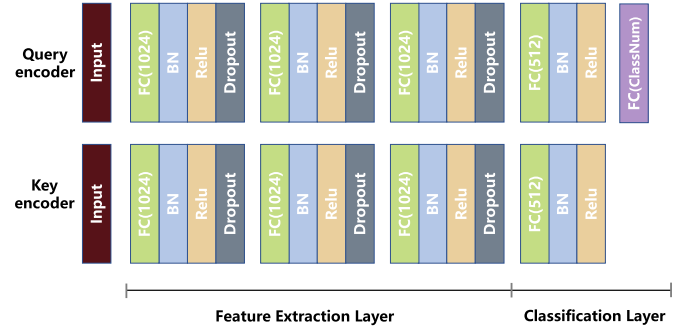


Fig. 4. The specific network structure of the M3SPCL.

5.3. Experiment settings

For all experiments, we adopt a unified data partition protocol by splitting each dataset into training, validation, and test subsets with a fixed ratio of 7:2:1. All compared methods are trained and evaluated under an identical random seed (10) for reproducibility. Baseline hyperparameters are set according to the official implementations or the configurations reported in the original papers, ensuring a consistent and fair experimental comparison.

For M3SPCL, it comprises V query encoders, key encoders, and classifiers. The encoders of query and key have the siamese structure $D^v - 1024 - 1024 - 512 - d$, where D^v and d are the dimensions of input and output space, respectively. Each encoder consists of fully connected networks, batch normalization layers, ReLU activation layers, and Dropout layers. Details are shown in Fig. 4. For parameter settings, the epoch is set to 120.

The Adam optimizer is used with a learning rate of 0.001 and a weight decay of 0.05 across all datasets. Batch size is selected from {32, 64, 128, 256, 512, 1024}. In addition, the parameter α is set to 0.1 for Caltech101, 0.5 for Fashion and Handwritten, and 0.2 for others. Grid search (Tang et al., 2024) is applied to select γ from {2, 3, 4, 5, 6} and τ from {0.05, 0.07, 0.09, 0.1}. The momentum coefficient μ is set within {0.7, 0.8, 0.9, 0.99, 0.999} to avoid drastic changes in prototypes during updates.

Table 4

Evaluation of the classification performance. Improvement denotes the gap between M3SPCL and the best benchmark methods.

Metric(%)	Method	Handwritten	LandUse21	BDGP	Scene15	Caltech101	Fashion	Reuters	NoisyMNIST	AVG.	AVG.Rank
Accuracy	DeepCon	97.917	68.403	98.580	50.781	45.170	96.637	76.961	95.565	78.752	7.000
	MvCLN	94.400	65.000	92.200	57.623	46.689	91.600	81.784	96.500	78.225	7.750
	DCP	97.800	75.050	96.880	75.340	54.686	97.390	82.000	98.160	84.663	6.000
	TMC	98.500	50.400	93.800	67.827	69.573	95.700	64.990	94.950	79.468	7.500
	ECML	99.400	53.917	97.333	76.190	71.847	95.550	65.821	90.233	81.286	6.625
	MTS-Net	98.568	77.827	99.740	81.959	74.978	99.169	88.368	99.320	89.991	3.375
	MvNNcor	99.479	82.031	99.716	82.813	76.344	98.268	89.283	98.694	90.828	3.000
	MvESR	99.740	82.552	99.574	82.366	76.786	98.300	89.025	99.296	90.955	2.750
	M3SPCL	99.826	87.109	99.858	84.668	78.164	99.355	90.215	99.498	92.337	1.000
	Improvement	0.087	4.557	0.118	1.856	1.378	0.186	0.932	0.178	1.162	-
Precision	DeepCon	98.313	66.818	98.643	46.069	15.467	96.735	48.762	95.557	70.795	6.750
	MvCLN	94.348	68.081	92.341	54.167	16.389	91.542	76.847	91.722	73.180	7.500
	DCP	97.794	75.000	96.906	74.454	41.996	98.716	78.898	98.134	82.737	5.000
	TMC	97.090	47.124	93.816	69.698	51.406	95.899	57.853	94.859	75.968	6.875
	ECML	98.345	53.010	97.257	67.900	53.238	95.615	65.077	90.379	77.603	6.250
	MTS-Net	99.500	75.344	95.833	67.413	48.457	86.525	75.695	82.979	78.968	6.500
	MvNNcor	99.648	83.683	99.705	82.808	61.067	98.395	87.749	98.745	88.975	2.750
	MvESR	99.862	84.330	99.561	81.784	60.184	98.959	87.992	98.982	88.957	2.375
	M3SPCL	99.877	87.938	99.773	85.100	65.657	99.321	90.137	99.529	90.916	1.000
	Improvement	0.015	3.608	0.068	2.292	4.590	0.361	2.145	0.547	1.703	-
Recall	DeepCon	98.112	69.930	98.598	49.627	16.822	96.664	38.371	95.532	70.457	6.500
	MvCLN	94.515	65.108	92.197	52.808	18.721	91.255	71.328	90.261	72.024	7.750
	DCP	97.696	74.050	96.794	74.630	33.413	98.400	81.338	98.146	81.808	5.000
	TMC	97.040	51.084	93.630	64.650	52.700	95.549	50.762	94.834	75.031	7.125
	ECML	98.495	56.655	97.607	68.412	54.763	95.498	52.506	90.035	76.746	6.125
	MTS-Net	98.625	69.887	97.088	65.253	40.770	86.068	72.977	82.670	76.667	6.375
	MvNNcor	99.630	81.667	99.747	82.373	61.490	98.264	86.889	98.719	88.597	2.750
	MvESR	99.862	82.997	99.633	82.082	61.141	98.933	87.191	98.777	88.827	2.375
	M3SPCL	99.889	86.045	99.899	82.080	65.474	99.312	87.789	99.505	90.219	1.000
	Improvement	0.027	3.048	0.152	1.467	3.984	0.380	0.598	0.727	1.298	-
Fscore	DeepCon	97.934	65.622	98.547	46.693	15.085	96.281	39.667	95.498	69.416	6.500
	MvCLN	94.369	61.046	92.226	49.686	14.131	91.036	73.279	89.880	70.707	7.875
	DCP	97.702	73.800	96.810	73.948	31.418	97.502	79.782	98.136	81.137	5.000
	TMC	96.945	44.434	93.574	62.903	49.486	95.328	49.596	94.532	73.350	7.250
	ECML	98.395	59.550	97.340	64.490	51.889	95.421	52.617	89.861	76.195	6.125
	MTS-Net	99.000	71.889	96.322	65.973	43.178	86.290	74.223	82.823	77.462	6.250
	MvNNcor	99.602	80.308	99.716	80.410	60.324	98.201	86.275	98.568	87.925	2.625
	MvESR	99.800	81.463	99.563	79.719	59.741	98.961	87.028	98.600	88.109	2.375
	M3SPCL	99.877	85.618	99.827	83.985	63.579	99.225	88.739	99.467	90.040	1.000
	Improvement	0.077	4.155	0.112	3.575	3.255	0.263	1.711	0.867	1.752	-

For baselines methods, **DeepCon** has one encoder with the architecture in $D^v - 1024 - 1024 - 512 - d$, where $d = 128$ is the dimensions of input and output. We concatenate the raw data as the input of **DeepCon**. For **MvCLN**, we fix the aligned proportion at 0.1, and then evaluate all pairwise combinations of views and use the best result for comparison. The ratio of negatives and positives is fixed at 30, and the switching time is set to 1. For **DCP**, the missing rate is set to 0 and the model uses the complete graph operation (Lin et al., 2022) for data with more than two views. Following the recommended parameter settings, we set the trade-off parameters α , λ_1 and λ_2 of **DCP** to 9, 0.1, and 0.1, respectively. In the experiments with **TMC** and **ECML**, we use the provided fully connected networks with an L2 regularization trade-off parameter of 0.0001 and a fixed lambda epoch value of 50. The temperature parameter τ of **MTS-Net** is selected from $\{1, 10, 100\}$, and γ of **MvNNcor** varies from 1.5 to 6 with the step of 0.5. The dimension of the output layer of **MvESR** is assigned to 128.

5.4. Classification results and analysis

Table 4 presents the performance in Accuracy, Precision, Recall and Fscore of all the methods across eight datasets. The best performances are highlighted in bold. From Table 4, we can observe that: 1) Across all datasets, M3SPCL consistently outperforms the competing methods, securing the top position. Our methods can surpass the best competitors

by 1.162%, 1.703%, 1.298% and 1.752% in the average of Accuracy, Precision, Recall, and Fscore, respectively. Notably, MvESR achieves the second position, demonstrating its strong performance as well. 2) For the **fusion strategy**, late fusion methods tend to exhibit better performance than the other methods. Furthermore, our method, M3SPCL, employs a multi-stage fusion strategy that integrates information at the early, intermediate, and late stages, achieving the best performance among all compared methods. 3) For the **utilization of principles**, it seems like methods implementing both consistency and complementarity principles are superior to those relying on a single principle alone. Among these methods, M3SPCL showcases the best performance, underscoring the importance of leveraging consistent and complementary information across different stages at multiple granularities. This advantage is particularly evident on complex datasets, such as Caltech101 with 6 views and 102 classes, where M3SPCL achieves a Precision improvement of 4.590% and a Recall improvement of 3.984%. 4) For the **correlation types**, consider the performance of CL-based methods including MvCLN and DCP. The primary distinction between these two methods lies in their treatment of correlations. To be specific, MvCLN focuses solely on category-level correlations, whereas DCP incorporates dual correlations, considering both category-level and instance-level correlations. Experimental results demonstrate that DCP consistently outperforms MvCLN across all evaluation metrics, emphasizing the benefit of leveraging dual correlations. Furthermore, our method, which also leverages dual cor-

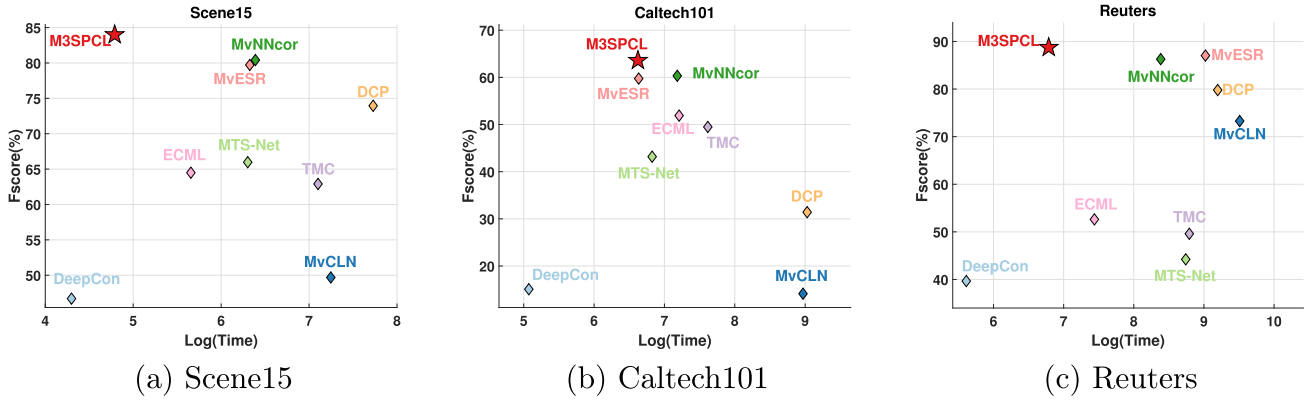


Fig. 5. Running time and Fscore of different methods. The horizontal axis is the natural logarithm value of the time (sec.) spent, and the vertical axis identifies the Fscore values for each method. The best results are signified by stars.

relations, surpasses DCP in performance. The key difference between our approach and DCP is the integration of a multi-stage fusion strategy, along with the additional implementation of the complementarity principle via both supervised CL and prototypical CL. This design allows our model to more effectively preserve both data commonality and diversity, thereby facilitating a more comprehensive exploitation of dual correlations and achieving superior performance.

5.5. Time analysis

In this section, we present the running time and Fscore of all methods on Scene15, Caltech101 and Reuters datasets to compare their computational efficiency and classification performance. In Fig. 5, methods closer to the top-left corner are deemed more effective and time-efficient. Obviously, our model M3SPCL demonstrates superior classification results and time-saving capabilities. Although the DeepCon architecture facilitates fast execution, its simplicity ignores the consistency across views, resulting in subpar performance.

Our approach enhances computational efficiency through the following mechanisms: 1) M3SPCL incorporates a paired view selection module, enabling it to process all views simultaneously without necessitating full graph iteration, as required by other CL-based methods. 2) Instead of focusing on instance-level correlations, our CL module directly consolidates consistent and complementary information at the category level, bypassing the need for complex representation matrix operations for each instance.

5.6. Performance with varying numbers of views

To examine the impact of the number of views on model performance, we conduct additional experiments on three datasets with the highest view counts: Caltech101 (6 views), Reuters (5 views), and Handwritten (5 views). In these experiments, we incrementally increase the number of views and evaluate the corresponding classification performance. The results, as shown in Fig. 6, indicate that incorporating additional views consistently enhances classification performance. This improvement can be attributed to the richer feature representations and more comprehensive multi-view correlations, which enable the model to more effectively capture the underlying data structures. Importantly, our model consistently outperforms the best competitor, MvESR, in both classification metrics and time efficiency. Furthermore, the performance advantage becomes increasingly pronounced as the number of views increases, underscoring the effectiveness of our approach in leveraging multi-view information.

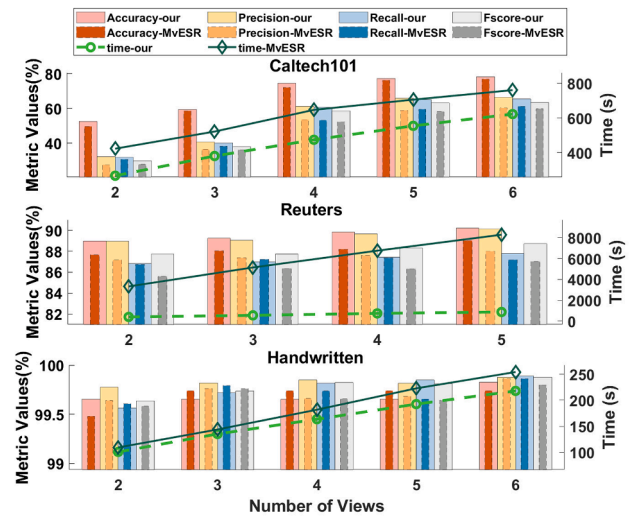


Fig. 6. Classification performance of our and the best benchmark (MvESR) under various views.

5.7. Ablation study

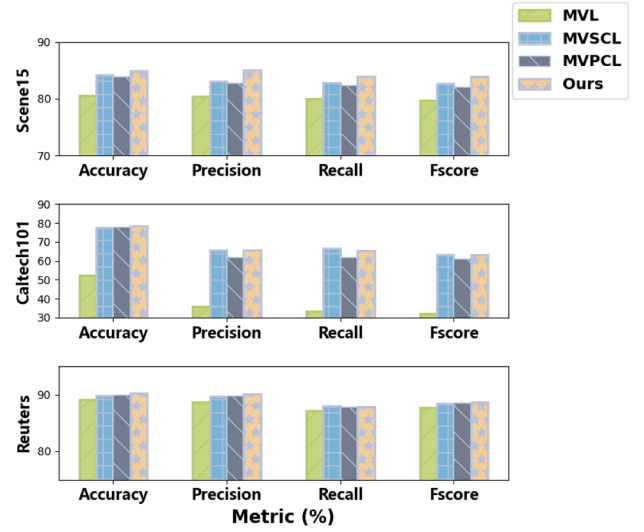
5.7.1. Module ablation

We perform an ablation study on all datasets to investigate the impact of different combinations of modules. In Type 0, we substitute the adaptive weight fusion strategy with average pooling in the decision fusion module, assuming equal importance for each view. In Type 1, the CL module is omitted, and classifiers are trained solely using view-specific representations. In Type 2, we employ the full graph paradigm to iterate through all views, as described in Fig. 3. As detailed in Table 5, the variations (Types 0, 1, and 2) exhibit different degrees of performance decline compared to M3SPCL, highlighting that all three modules in our proposed method contribute to classification performance, with varying levels of impact. Concretely, the most pronounced performance degradation in M3SPCL occurs when the CL module is omitted (Type 1), underscoring the essential role of the CL module in facilitating the integration of consistent information alongside complementarity. The second most significant performance decline is observed when all views are treated equally (Type 0), highlighting the critical importance of incorporating adaptive decision-making to account for the relative significance of different views. In contrast, the Type 2 method experiences the least performance drop, which can be attributed to the ability of the full graph CL to effectively capture information across all possible view pairs, even in the absence of the PVSM mechanism.

Table 5

Classification performance (%) on the three modules (The bold values are the best result, the numbers in parentheses represent the rank, and the ✓ denotes the method with the component).

Type	Components			Handwritten			LandUse21			BDGP			Scene15		
	PVSM	CLM	DFM	Accuracy	Precision	Recall	Fscore	Accuracy	Precision	Recall	Fscore	Accuracy	Precision	Recall	Fscore
0	✓	✓		99.610	99.675	99.625	99.633	83.399	84.773	84.763	82.695	99.609	99.595	99.625	99.605
1	✓		✓	99.219	99.290	99.240	99.220	82.986	83.169	83.472	81.016	99.574	99.605	99.616	99.596
2		✓	✓	99.610	99.680	99.630	99.620	84.641	85.500	84.810	84.440	99.609	99.630	99.640	99.630
our	✓	✓	✓	99.826	99.877	99.889	99.877	87.109	87.938	86.045	85.618	99.858	99.773	99.899	99.827
Type	Components			Caltech101			Fashion			Reuters			NoisyMNIST		
	PVSM	CLM	DFM	Accuracy	Precision	Recall	Fscore	Accuracy	Precision	Recall	Fscore	Accuracy	Precision	Recall	Fscore
0	✓	✓		76.578	67.069	63.241	63.527	98.947	99.002	98.991	98.959	89.732	89.531	87.594	88.220
1	✓		✓	52.461	35.678	33.226	32.004	98.981	99.136	99.047	98.940	89.121	88.615	87.130	87.719
2		✓	✓	77.391	69.186	63.690	63.532	99.102	99.106	99.132	99.110	90.030	89.530	88.196	88.613
our	✓		✓	78.164	65.657	65.474	63.579	99.355	99.321	99.313	99.225	90.215	90.137	87.789	88.739

**Fig. 7.** Ablation study on different loss functions.

5.7.2. Loss ablation

To further investigate the impact of different combinations of CL losses on classification performance, we randomly select three datasets representing distinct sample size ranges: below 5000 (Scene15), between 5000 and 10,000 (Caltech101), and above 10,000 (Reuters). The performance of different loss combinations on these three datasets is demonstrated in Table 6. The MVL method, which does not incorporate any CL loss, demonstrates the poorest performance due to its inability to capture consistent information, highlighting the necessity of incorporating contrastive learning losses. Building on MVL, MVPCL and MVSCL introduce different types of CL losses, resulting in improved performance. MVPCL, which focuses on category-level complementarity through prototypical contrast, outperforms MVSCL by leveraging inter-class relationships. However, its reliance solely on prototypes leads to information loss and overlooks intra-class diversity, resulting in suboptimal performance compared to M3SPCL. In contrast, M3SPCL, which integrates both $\mathcal{L}_{scl}$ and $\mathcal{L}_{pcl}$, achieves the best performance across all datasets, with particularly notable improvements observed in Scene15 and Caltech101. This is attributed to its ability to comprehensively capture both intra- and inter-class information. These findings underscore not only the individual contributions of $\mathcal{L}_{scl}$ and $\mathcal{L}_{pcl}$ but also their synergistic effect in enhancing classification performance Fig. 7.

5.7.3. Correlation ablation

In this ablation study, we compare the model's performance with and without category-level correlations to assess their impact. Specifically, the setting "w/ category-level correlations" represents dual correlations, which include both category-level and instance-level information, while the setting "w/o category-level correlations" considers only instance-level correlations. The radar chart results in Fig. 8 reveal that incorporating category-level correlations consistently improves performance across multiple datasets. Moreover, the performance gap becomes increasingly pronounced as the number of categories grows, underscoring the effectiveness of category-level correlations in fully leveraging multi-view information and enhancing model performance.

5.8. Parameter analysis

Fig. 9 illustrates the parameter sensitivities of the proposed method in three datasets. Taking the Fscore value as an example, there are some underlying conclusions as follows:

1) The optimal value of γ varies across datasets: $\gamma = 3$ for Scene15, $\gamma = 2$ for Caltech101, and $\gamma = 5$ for Reuters. These results highlight the

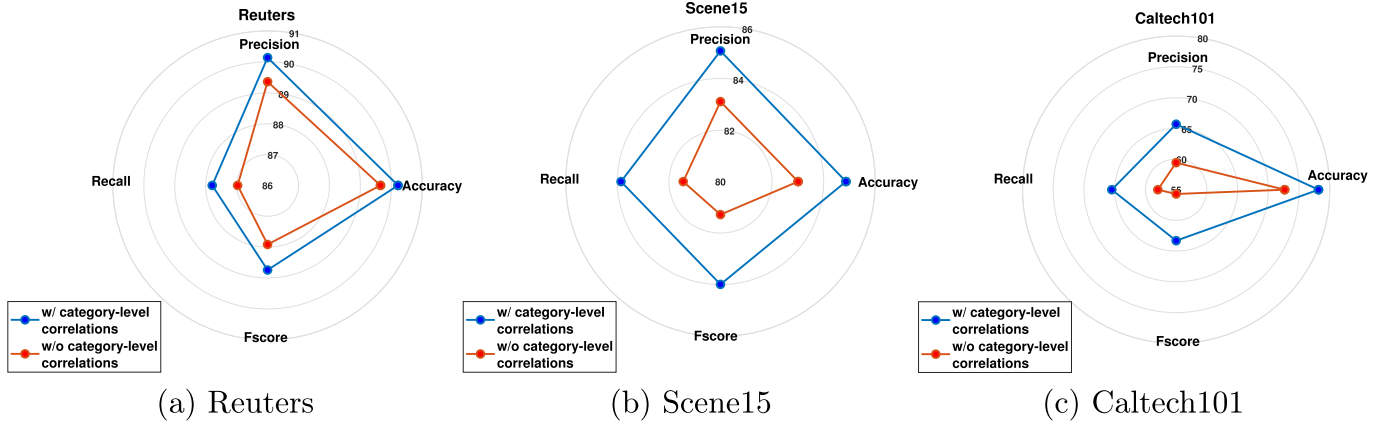


Fig. 8. Performance of M3SPCL with and without category-level correlations on Reuters (5 categories), Scene15 (15 categories), and Caltech101 (101 categories).

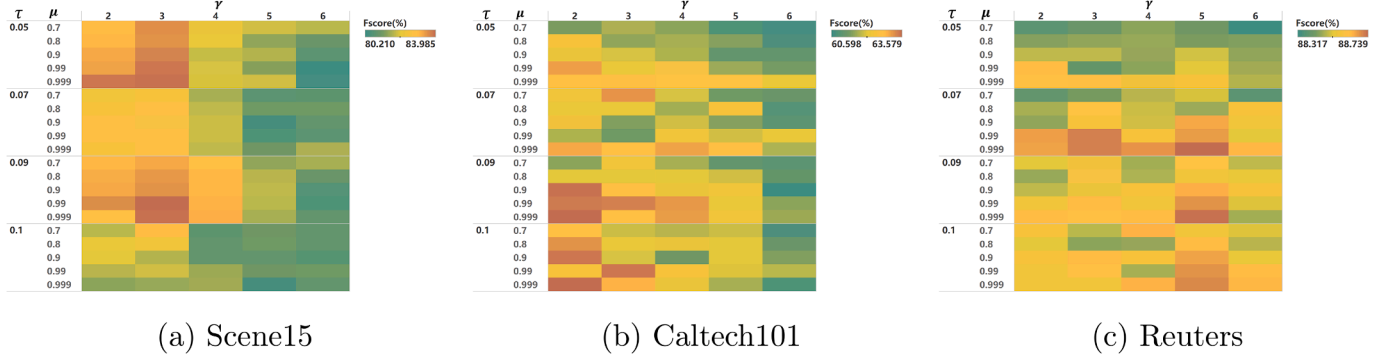


Fig. 9. Parameters sensitivity analysis of Fscore value on Scene15, Caltech101 and Reuters.

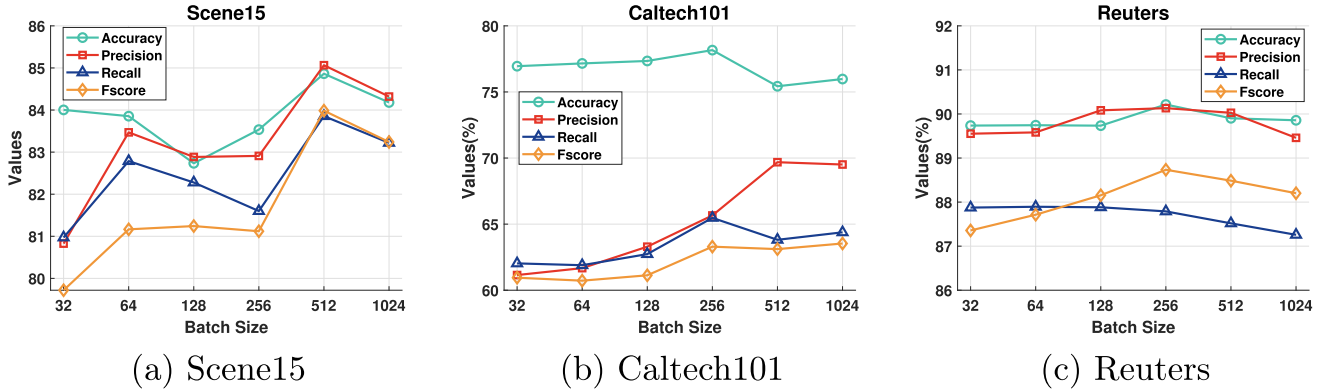


Fig. 10. Results of M3SPCL in various datasets with different batch sizes.

dataset-specific nature of γ and demonstrate the importance of carefully tuning it to capture the unique characteristics of each dataset.

2) Higher τ values increase the softmax output variance, resulting in a stronger penalty, which is beneficial for more complex datasets with many samples and views (e.g., Caltech101 and Reuters). However, for datasets with fewer views, such as Scene15, smaller τ values are more appropriate, as excessively large τ can lead to overfitting.

3) Momentum parameter μ controls the variation degree of the prototypes. The proposed method performs better with bigger μ values in different datasets. This suggests that a relatively larger momentum helps stabilize the updating process and contributes to learning compact representations.

Furthermore, we investigate the effect of batch size variations on the performance of M3SPCL, as it is widely recognized as a critical factor influencing contrastive learning performance (He et al., 2020a; Tian et al.,

2020). As shown in Fig. 10, the metrics across different domains generally increase with batch size up to a certain point, after which they exhibit a slight decline. Specifically, the optimal results are achieved with a batch size of 512 in Scene15 and 256 in both Caltech101 and Reuters. While these values may not be considered small in a general sense, they are relatively modest compared to the large batch sizes often required in contrastive learning settings. Based on these findings, we recommend using moderate batch sizes (e.g., 256-512) when applying M3SPCL, as this range balances computational efficiency and model performance.

5.9. Visualization of features

To further validate the effectiveness of the proposed method, we visualize the embedding feature spaces learned by M3SPCL using T-SNE

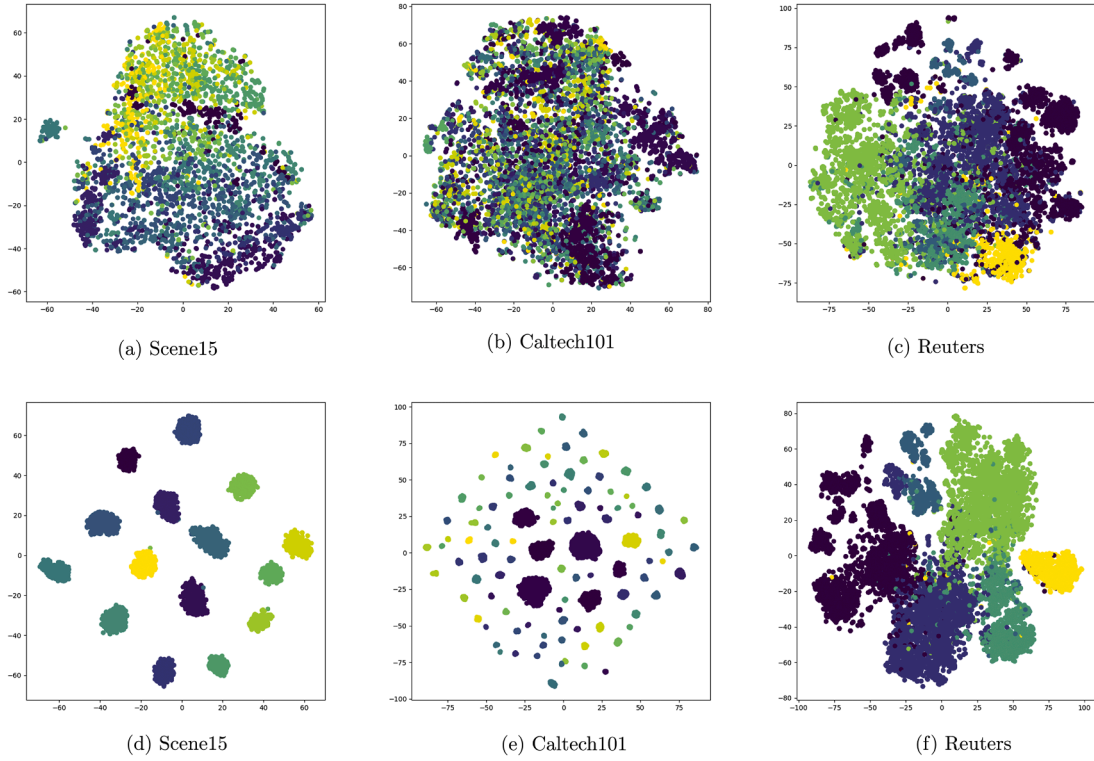


Fig. 11. T-SNE visualization. The top row shows the raw features at the initial epoch, while the bottom row displays the learned representations at the final epoch.

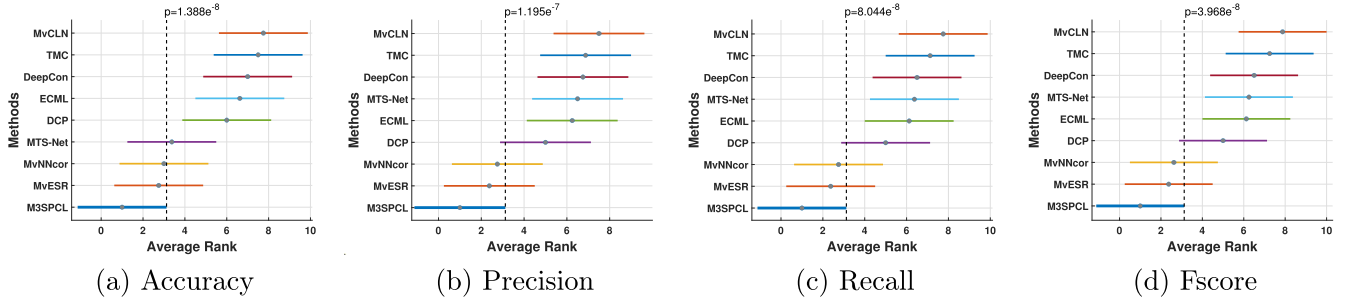


Fig. 12. The y-axis lists all methods. The x-axis shows the average rank. A smaller rank means better performance. Each method is represented by a horizontal segment. The midpoint is its average rank. The segment length corresponds to CD. If two segments do not overlap, their difference is statistically significant.

on the aforementioned three datasets. Fig. 11 illustrates the raw features (top row) and the learned fusion representations (bottom row) of these datasets. The visualized results show that the representations learned by M3SPCL are both compact and separable, validating its effectiveness across different datasets.

5.10. Non-parametric statistical analysis

We conduct a non-parametric statistical analysis to compare different models. Specifically, we employ the Friedman test followed by the Nemenyi post-hoc procedure on four evaluation metrics, including Accuracy, Precision, Recall, and Fscore. For all metrics, the Friedman test yields p-values smaller than 0.05, indicating that the null hypothesis should be rejected and that statistically significant performance differences exist among the compared methods.

To further identify pairwise differences, we apply the Nemenyi post-hoc test. It computes the critical difference (CD), which serves as a threshold for determining whether the rank difference between two methods is statistically significant. The CD is calculated as

$$CD = q_\alpha \sqrt{\frac{D(D+1)}{6T}}, \quad (13)$$

where q_α is the critical value derived from the Tukey distribution, T denotes the number of datasets, and D is the number of compared methods. In our experiments, we obtain $CD = 1.8699$ under $D = 9$, $T = 8$, and $\alpha = 0.05$. As shown in Fig. 12, M3SPCL achieves the best (lowest) average rank and consistently ranks first across all metrics. It also shows significant advantages over most baseline methods. These results confirm the effectiveness and robustness of M3SPCL.

Table 6
Classification performance (%) on different loss functions (The bold values are the best results).

Type	Component		Scene15		Caltech101				Reuters			
	L_{ce}	$L_{s,d}$	L_{pcl}	Accuracy	Precision	Recall	Fscore	Accuracy	Precision	Recall	Fscore	Fscore
MVL	✓			80.566	80.465	80.030	79.670	52.461	35.678	33.226	32.004	87.719
MVSL	✓	✓		83.984	82.790	82.435	82.030	77.753	61.837	62.112	60.964	88.457
MVPL	✓			84.180	83.045	82.810	82.650	77.422	65.930	66.490	63.292	88.560
M3SPCL	✓	✓	✓	84.668	85.100	83.840	83.985	78.164	65.657	65.474	63.579	88.739

6. Conclusion and future work

In this paper, we propose M3SPCL, a multi-view learning framework that aims to balance effectiveness and efficiency by jointly modeling consistency and complementarity. Extensive experiments on multiple multi-view benchmarks demonstrate that M3SPCL consistently outperforms state-of-the-art methods in both predictive performance and runtime efficiency. From an effectiveness perspective, experimental results on eight benchmark datasets show that M3SPCL achieves average improvements of over 1.162%, 1.703%, 1.298%, and 1.752% in Accuracy, Precision, Recall, and F-score, respectively. Ablation studies further confirm that each fusion stage contributes complementary gains, and that the dual-level contrastive learning mechanism is critical for capturing semantic consistency beyond instance-level alignment. From an efficiency perspective, runtime analysis indicates that M3SPCL achieves a favorable balance between performance and computational cost. Notably, as the number of views increases, the efficiency advantage of M3SPCL becomes increasingly evident, while consistently maintaining superior classification performance over strong baselines.

In future work, an important direction is to extend M3SPCL to more challenging multi-view scenarios involving high heterogeneity or noisy views, where robustness and stability under uncertainty become critical. While the current framework focuses on balancing effectiveness and efficiency through dual correlations, future extensions could incorporate robustness-aware mechanisms, such as modeling multi-modal category distributions with multiple prototypes or introducing uncertainty-aware view fusion to better assess view reliability.

CRedit authorship contribution statement

Jingjing Tang: Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Funding acquisition; **Yan Li:** Writing – original draft, Validation, Methodology, Investigation, Funding acquisition, Formal analysis; **Saiji Fu:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization; **Yingjie Tian:** Writing – review & editing, Supervision, Conceptualization; **Yu Bao:** Writing – review & editing, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Proof of the weight updating function

To optimize problem (10), we can obtain the Lagrangian formula as:

$$\mathfrak{L}(m^v, \lambda) = \sum_{v=1}^V (m^v)^\gamma \mathcal{L}^v(Q, K, P, Y, S) + \lambda \left(1 - \sum_{v=1}^V m^v \right), \quad (\text{A.1})$$

where λ is the Lagrangian multiplier. By calculating the partial derivatives of Formula (A.1) with respect to m^v and λ , and setting them to zero, we can obtain:

$$\frac{\partial \mathfrak{L}}{\partial m^v} = \gamma (m^v)^{\gamma-1} \mathcal{L}^v - \lambda = 0; \quad \frac{\partial \mathfrak{L}}{\partial \lambda} = \sum_{v=1}^V m^v - 1 = 0. \quad (\text{A.2})$$

Then, there is

$$m^v = \left(\frac{\lambda}{\gamma \mathcal{L}^v} \right)^{\frac{1}{\gamma-1}}. \quad (\text{A.3})$$

Substitute (A.3) to $\sum_{v=1}^V m^v = 1$, we can get:

$$\sum_{v=1}^V m^v = \sum_{v=1}^V \left(\frac{\lambda}{\gamma \mathcal{L}^v} \right)^{\frac{1}{\gamma-1}} = \left(\frac{\lambda}{\gamma} \right)^{\frac{1}{\gamma-1}} \sum_{v=1}^V (\mathcal{L}^v)^{\frac{1}{1-\gamma}} = 1. \quad (\text{A.4})$$

Then we derive $\left(\frac{\lambda}{\gamma} \right)^{\frac{1}{\gamma-1}} = \frac{1}{\sum_{v=1}^V (\mathcal{L}^v)^{\frac{1}{1-\gamma}}}$, and substituting it into Formula (A.3), we obtain:

$$m^v = \frac{(\mathcal{L}^v)^{\frac{1}{1-\gamma}}}{\sum_{l=1}^V (\mathcal{L}^l)^{\frac{1}{1-\gamma}}}. \quad (\text{A.5})$$

Appendix B. Lower bound of the term (b) of the proposed CL loss

Suppose all training samples are accessible, denoted as $|A(x_i)| = N$, with a minimum of two categories ($C \geq 2$). According to Jensen's inequality, term (b) can be derived as:

$$\begin{aligned} \frac{1}{N} \log \prod_{i=1}^N A_i^v B_i^v &= \frac{1}{N} \log \prod_{i=1}^N \sum_{a \in A(x_i)} \exp \left(\frac{q_i^v \cdot k_a^v}{\tau} \right) \sum_{j=1}^C \exp \left(\frac{q_i^v \cdot p_j^v}{\tau} \right) \\ &= \frac{1}{N} \log \prod_{i=1}^N N \sum_{a \in A(x_i)} \frac{1}{N} \exp \left(\frac{q_i^v \cdot k_a^v}{\tau} \right) C \sum_{j=1}^C \frac{1}{C} \exp \left(\frac{q_i^v \cdot p_j^v}{\tau} \right) \\ &\geq \frac{1}{N} \log \prod_{i=1}^N \left[N \exp \left(\frac{1}{N} \sum_{a \in A(x_i)} \frac{q_i^v \cdot k_a^v}{\tau} \right) C \exp \left(\frac{1}{C} \sum_{j=1}^C \frac{q_i^v \cdot p_j^v}{\tau} \right) \right] \\ &= \log(NC) + \frac{1}{N} \log \prod_{i=1}^N \exp \left(\frac{1}{N\tau} \sum_{a \in A(x_i)} q_i^v \cdot k_a^v + \frac{1}{C\tau} \sum_{j=1}^C q_i^v \cdot p_j^v \right) \\ &= \log(NC) + \frac{1}{N\tau} \sum_{i=1}^N \left(\frac{1}{N} \sum_{a \in A(x_i)} q_i^v \cdot k_a^v + \frac{1}{C} \sum_{j=1}^C q_i^v \cdot p_j^v \right) \\ &\geq \log(NC) + \frac{1}{N\tau} \sum_{i=1}^N \left(\frac{1}{N} (-N) + \frac{1}{C} (-C) \right) \\ &= \log(NC) - \frac{2}{\tau}. \end{aligned}$$

Due to $-1 \leq q \cdot k \leq 1$, the term (b) has the lower bound value $\log(NC) - \frac{2}{\tau}$.

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